Design Decisions and Trade-offs

Project 2: Startup Screening Workflow. Deterministic scoring on AWS Bedrock Flows.

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Each entry records a decision made while building the reference implementation, the reasoning behind it, and the alternative that was considered and rejected.

1. Orchestration: Bedrock Flows over an autonomous agent

Decision: Build the screener as a Bedrock Flow, a deterministic orchestrated workflow, rather than an autonomous agent.

Rationale: The original plan targeted Bedrock Agents, but that service entered maintenance mode and closed to new customers, so building on it would have shipped a portfolio piece on a retiring service. Of the current paths, AgentCore is code-forward and heavy for this scope, while Bedrock Flows is low-code and fits the task. More importantly, a screening-and-scoring task benefits from determinism: repeatable, auditable, and testable results matter more here than autonomous tool use.

Rejected: An autonomous agent (AgentCore or Agents Classic). It would have added engineering and non-determinism for no benefit to a scoring task, and the classic path was closing regardless.

2. Cloud and region: AWS, Europe (Frankfurt)

Decision: Build on AWS in Europe (Frankfurt), eu-central-1.

Rationale: AWS demonstrates a second cloud alongside the Azure build in Project 1. Frankfurt keeps the work inside the EU, consistent with the EU-residency posture that runs through the portfolio, and it carries the broadest Bedrock model and Flows feature support of the EU regions.

Rejected: A US region, which would have been simpler for model availability but would have broken the EU-residency thread. Ireland was a valid EU alternative; Frankfurt was chosen for feature breadth.

3. Model: Claude Sonnet 4.5, a mid-tier reasoning model

Decision: Use Claude Sonnet 4.5 as the scoring model.

Rationale: Scoring a startup against a rubric needs solid reasoning but not a top-cost model, mirroring the mini-tier logic from Project 1. Sonnet 4.5 gave accurate, well-justified scores that discriminated cleanly across the range in testing.

Rejected: A frontier-tier model, on cost, since the task did not require it. Amazon Nova was considered as a Marketplace-free alternative and remains a valid fallback; Sonnet was chosen for reasoning quality once access was cleared.

4. Scoring design: six weighted dimensions, 1 to 5, banded

Decision: Score against six weighted dimensions (Team and Traction at weight 3; Market, Differentiation, and Thesis fit at weight 2; Investment readiness at weight 1), each 1 to 5, summed to a weighted total out of 65 and banded Advance, Borderline, or Pass.

Rationale: These are the real dimensions used to screen startups for investor matching, which makes the tool defensible rather than generic. The weights reflect genuine screening priorities, and the banding turns a raw score into an actionable recommendation.

Rejected: An unweighted or simple average, which would have treated all dimensions as equally important and lost the real priority order. A pass/fail-only output was rejected for hiding the reasoning behind the call.

5. Output format: human-readable prose

Decision: Return the assessment as clean prose, a score and one-line justification per dimension, then the total and recommendation.

Rationale: The screener supports a human-in-the-loop decision, so the output is written for a person reviewing applications, not for a machine. Readability at a glance is the right design for that reader, and it fits the governance principle that screening feeds a human, not an automated reject.

Rejected: Structured JSON output. It is easier to score programmatically, but it optimizes for a machine consumer this tool does not have, and it reads worse for the human reviewer who is the actual user.

6. PII handling: mask on both input and output

Decision: Attach a Bedrock Guardrail that detects PII (name, email, phone, address) and masks it on both ingress and egress.

Rationale: The screener evaluates the business, not the people, so founders' personal details should never reach the model in the clear. Masking on input keeps PII out of processing; masking on output is defense in depth in case anything is echoed or missed. This is privacy-by-design and pairs with the least-privilege and EU-residency posture from Project 1.

Rejected: Blocking on PII, which would reject a valid submission just for naming a founder, too aggressive for this tool. Masking one direction only was rejected as leaving a gap on the other.

7. Routing: condition node considered and deliberately not shipped

Decision: Considered a condition node to route on the recommendation band, then chose to keep the screener lean without it.

Rationale: Reliable numeric branching required extracting a structured value from the model's text output, which the workflow surfaces as a string, not parsed data. Making the branch robust would have meant adding a parsing step purely to support one routing node. For a reference implementation, that complexity was not justified.

Rejected: Adding an inline-code parser to force the branch to resolve. It would have worked but over-engineered a portfolio piece, exactly the kind of complexity that does not earn its place. A production version would emit structured output natively and route on it.

8. Validation: a ten-case evaluation harness

Decision: Test the screener against ten startup cases spanning the range (three expected Advance, four Borderline, three Pass), comparing each actual band to an expected band set in advance.

Rationale: An eval turns "it seems to work" into measured evidence, which is the LLMOps discipline most portfolios skip. Setting expected bands before running removes hindsight bias, and the spread deliberately includes borderline and buzzword-heavy cases to test discrimination.

Rejected: Ad hoc testing on a few favourable inputs, which would have proved nothing about consistency or edge behaviour. Automating the batch run via the API was rejected as unnecessary engineering for the signal it adds at this scope.

Note on scope

The guardrail and PII details also feed the security and governance note for this project; here decision 6 states the choice, and the governance note expands the full data-path posture.